
CLUSTER

System Architecture Document

Stablecoin Rail Infrastructure — Naira ↔ USDT/USDC Multi-chain

Version 2.0 — Including Backend Upgrade Strategy

Confidential — Internal Use Only

May 2026

1. System Overview

Clusteer is a stablecoin rail platform enabling Nigerian users to buy, sell, send, receive, and hold USDT and USDC across four blockchain networks (BSC, Solana, TRON, Ethereum). The platform provides custodial wallets, real-time exchange rates, bank-grade KYC, and fiat on/off ramps via Naira (NGN).

Technology Stack

Layer	Technology	Version
Frontend	Next.js + React + TypeScript	16.1.1 / 19.2.3
API Gateway	Next.js API Routes (Proxy)	16.1.1
Backend API	Spring Boot (Java)	3.3.3 / Java 21
Blockchain Engine	Django + DRF (Python)	5.1.10 / Python 3.10
Database	PostgreSQL	14+
Cache / Queue	Redis	7+
Task Queue	Celery + Celery Beat	5.5.3
Auth Provider	Firebase Auth + Admin SDK	13.6.0
Frontend Hosting	Vercel	Latest
Backend Hosting	Railway	Docker-based

2. High-Level Architecture

2.1 Current Architecture

```
INTERNET → Vercel CDN → clusteer-unified (Next.js 16, SSR)
■
■ ■ API Proxy Routes (/api/*) ■■■■■■
■ ■
■ ■ Firebase Auth/Firestore ■■■■■■ X-API-KEY
■ (Auth, RTDB, Storage) ■
■ ▼
■ ■ Spring Boot API ■■■■■■ Django Blockchain Engine
■ (Railway, :8080) (Railway, :8000)
■ User mgmt, admin ■
■ ■ Celery Workers (sweep_bsc/sol/tron)
■ ■ Webhook handlers (/bsc/sol/tron/webhook/)
■ ■ Chain managers (web3, solana-py, tronpy)
■
■ ■ DATA LAYER
■ ■ PostgreSQL (primary DB, 2 instances)
■ ■ Redis (cache, Celery broker)
■ ■ Firebase (auth, Firestore, Storage)

BLOCKCHAIN LAYER
■ ■ BSC RPC (Binance DataSeed)
■ ■ Solana RPC (Mainnet Beta)
■ ■ TRON RPC (TronGrid)
■ ■ Tatum (deposit webhook subscriptions)
```

2.2 Target Architecture (Post-Upgrade)

```
INTERNET → Vercel CDN + Sentry SDK
■
■ ■ clusteer-unified (Next.js 16)
■ + Rate Limiting (Redis-backed)
■ + Idempotency Keys
■ + Request ID (UUID per request)
■ + HMAC Request Signing
■ + Structured Logging (pino)
■
■ ■ Firebase Auth/Firestore ■■■■■■
■ ■ HMAC-SHA256 Signed
■ ▼
■ ■ Spring Boot API Django Blockchain Engine
■ (Railway, :8080) (Railway, :8000)
■ + RBAC Admin Permissions
■ + SystemAuditLog (immutable)
■ + Sentry SDK
■ + Structured JSON Logs
■ ■
■ ■ ■ blockchain_worker:
■ ■ sweep, withdraw, confirm_tx
■ ■ ■ notification_worker:
■ ■ email (Resend), SMS (Termii), push (FCM)
■ ■ ■ maintenance_worker:
■ ■ reconcile, data_export, cleanup
■
■ ■ DATA LAYER
■ ■ PostgreSQL + PgBouncer + Read Replica
■ ■ (partitioned WalletTransaction by month)
■ ■ Redis (4 logical DBs):
■ ■ 0=Celery broker, 1=cache+rate, 2=realtime, 3=sessions
■ ■ Firebase (auth, Firestore, Storage)
```

BLOCKCHAIN LAYER

- ■ BSC RPCs (3 nodes, failover + circuit breaker)
- ■ Solana RPCs (3 nodes, failover)
- ■ TRON RPCs (3 nodes, failover)
- ■ Tatum (deposit webhooks, HMAC-verified)

OBSERVABILITY

- ■ Sentry (errors, Django + Next.js)
- ■ Prometheus / Datadog (metrics)
- ■ PagerDuty (critical alerts)
- ■ Structured JSON Logs → CloudWatch

3. Service Inventory

Service	Tech	Port	Deploy	Purpose
clusteer-unified	Next.js 16	3000	Vercel	Frontend + API proxy
Blockchain Engine	Django 5.1	8000	Railway	Wallets, blockchain, orders, KYC
Clusteer-API	Spring Boot 3.3	8080	Railway	User mgmt, admin ops
Celery Workers	Celery 5.5	—	Railway	Background tasks
PostgreSQL	PG 14+	5432	Railway	Primary database
Redis	Redis 7+	6379	Railway	Cache, broker, rate limiting
Firebase	v9	—	Google Cloud	Auth, Firestore, Storage

3.1 Service Communication

Source	Target	Protocol	Auth
Browser	Next.js	HTTPS	Firebase JWT (cookie)
Next.js	Django	HTTPS	X-API-KEY (+ HMAC planned)
Next.js	Spring Boot	HTTPS	X-API-KEY
Next.js	Firebase	HTTPS	Admin SDK
Django	BSC/SOL/TRON RPC	HTTPS	API key
Django	Smile Identity	HTTPS	HMAC-SHA256
Django	Quidax	HTTPS	API key
Tatum	Django (webhook)	HTTPS	HMAC-SHA256

4. Data Layer

4.1 Database Models (18 current + 7 planned)

Model	Purpose	Key Fields
MultiChainWallet	User wallet container	user_id (unique)
UserBlockchain	Per-chain address + encrypted key	chain_name, token_type, balance, private_key
FiatBalance	NGN balance	currency, balance
WalletTransaction	All tx records	tx_id (unique), direction, status, amount
Order	Buy/sell trades	order_type, status, exchange_rate, blockchain_tx_hash
OrderStatusHistory	Order audit trail	from_status, to_status, changed_by
KYCVerification	Identity records	bvn_encrypted, nin_encrypted, bvn_hash
KYCAuditLog	Compliance trail	action, performed_by, ip_address
BankAccount	Linked banks	bank_name, account_number, is_verified
SupportTicket	Support tickets	ticket_number, category, status
AccountLimits	Tx limits	daily_withdrawal_limit, monthly_usage

Planned Models (Upgrade Strategy)

Model	Purpose	Phase
IdempotencyRecord	Prevent duplicate transactions	P1
ReconciliationLog	DB vs on-chain balance tracking	P1
SystemAuditLog	Immutable audit trail	P3
AdminPermission	RBAC for admin	P2
PaymentTransaction	Paystack/Flutterwave records	P6
ReferralCode + ReferralReward	Referral system	P6

4.2 Redis Architecture

DB	Purpose	TTL
0	Celery task broker	Consumed on processing
1	Django cache + rate limiting	2-5 min (data), 1-60 min (rate)
2 [planned]	Realtime data (rates, prices)	30s-5min
3 [planned]	Admin sessions	8 hours

5. Blockchain Layer

Chain	RPC Provider	Tokens	Library
BSC	Binance DataSeed	USDT, USDC	web3.py 7.6.1
Solana	Mainnet Beta	USDT, USDC	solana 0.36.6 + solders
TRON	TronGrid	USDT, USDC	tronpy 0.5.0
Ethereum	TBD	USDT, USDC	web3.py (partial)

Smart Contract Addresses

Token	Chain	Address
USDT	BSC	0x55d398326f99059fF775485246999027B3197955
USDT	TRON	TR7NHqjeKQxGTCi8q8ZY4pL8otSzgJLj6t
USDT	Solana	Es9vMFrzaCERmJfrF4H2FYD4KCoNkY11McCe8BenwNYB

Wallet Model: Custodial — AES-256-GCM encrypted private keys, server-side.

Per user: 1 MultiChainWallet → 4 chains × 2 tokens = 8 UserBlockchain records + 1 FiatBalance (NGN).

Vault Sweeping Schedule

Task	Schedule	Action
sweep_bsc	2:00 AM UTC	Fund BNB gas → sweep USDT to vault
sweep_sol	2:20 AM UTC	Fund SOL gas → sweep USDT to vault
sweep_tron	2:40 AM UTC	Fund TRX gas → sweep USDT to vault

6. Security Architecture

6.1 Encryption Matrix

Data	Algorithm	Key
Custodial private keys	AES-256-GCM (Fernet)	CUSTODIAL_WALLET_ENCRYPTION_KEY
Vault private keys	AES-256-GCM (Fernet)	VAULT_WALLET_ENCRYPTION_KEY
BVN, NIN, phone, address	Fernet (AES-128-CBC)	DJANGO_ENCRYPTION_KEY
BVN/NIN dedup	SHA-256 (one-way hash)	No key
2FA pending tokens	HS256 (HMAC-JWT)	JWT_SECRET
Spring Boot columns	PBEWithMD5AndDES	COLUMN_ENCRYPTION_KEY

6.2 Network Security

Control	Status
HTTPS redirect + HSTS (1 year, preload)	Active
X-Frame-Options DENY	Active
Content-Type nosniff	Active
CORS restricted to allowed origins	Active
httpOnly + secure + sameSite cookies	Active
Webhook HMAC-SHA256 verification	Defined, NOT enforced (P0 fix)
HMAC request signing (Next.js → Django)	Planned (P2)
Rate limiting	In-memory active, Redis planned

7. External Integrations

Service	Provider	Purpose	Status
Auth & DB	Firebase	User auth, Firestore, Storage	Active
Blockchain Monitor	Tatum	Deposit webhooks	Active
Exchange	Quidax	Live rates, order execution	Active
Market Data	CoinGecko	Prices + sparklines	Active
Forex	Open ER API	USD/NGN rate	Active
KYC Primary	Smile Identity	BVN/NIN verification	Active
KYC Fallback	Youverify	BVN/NIN verification	Configured
Email	SendGrid	Transactional email	Configured
SMS	Termii	OTP + notifications	Configured
Payments	Paystack	NGN on/off ramp	Configured
Payments Alt	Flutterwave	NGN on/off ramp	Configured
Error Tracking	Sentry	Error monitoring	Pending DSN

8. API Surface Area (52 Routes)

8.1 Money Endpoints (Critical Path)

Route	Method	Purpose
/api/wallet	GET	Fetch wallet balances + addresses
/api/trade	POST	Buy/sell USDT
/api/transfer/send	POST	Withdraw crypto or NGN to bank
/api/transfer/internal	POST	P2P platform transfer
/api/order	GET	List trade orders
/api/order/[id]/cancel	POST	Cancel pending order
/api/swap	POST	Cross-chain swap
/api/withdrawal	POST	NGN bank withdrawal

8.2 Auth Endpoints

Route	Method	Auth
/api/auth-firebase/login	POST	Public (rate limited 5/min)
/api/auth-firebase/register	POST	Public (rate limited 5/min)
/api/auth-firebase/logout	POST	auth_token cookie
/api/auth-firebase/verify-2fa	POST	pending_2fa_token cookie
/api/auth-firebase/reset-password	POST	Public

9. Background Jobs

9.1 Current Tasks

Task	Schedule	Purpose
sweep_bsc_wallet_task	Daily 2:00 AM	Sweep BSC wallets to vault
sweep_sol_wallet_task	Daily 2:20 AM	Sweep Solana wallets to vault
sweep_tron_wallet_task	Daily 2:40 AM	Sweep TRON wallets to vault

9.2 Planned Tasks (Upgrade)

Task	Schedule	Worker	Phase
confirm_pending_transactions	Every 2 min	blockchain_worker	P1
reconcile_balances	Daily 3:00 AM	maintenance_worker	P1
send_email_notification	On-demand	notification_worker	P6
send_sms_notification	On-demand	notification_worker	P6
process_data_export	On-demand	maintenance_worker	P6
rpc_health_check	Every 30s	blockchain_worker	P5
cleanup_expired_idempotency	Daily 4:00 AM	maintenance_worker	P1

10. Monitoring & Observability

Critical Alerts

Alert	Condition	Severity
Vault balance low	< \$500 on any chain	Critical
Withdrawal stuck	Pending > 30 min	Urgent
Reconciliation mismatch	DB vs chain > 0.01 USDT	High
RPC endpoint down	All fallbacks failed	Critical
Sweep task failure	Celery exception	High
Auth brute force	> 10 failed/hour from 1 IP	Medium
Order failure spike	> 5 failed/hour	High
API latency	p95 > 2 seconds	Medium

11. Upgrade Roadmap

Phase	Name	Timeline	Key Deliverables
P0	Critical Blockers	Week 1	Real tx hashes, webhook HMAC, rate limiting, email verification
P1	Transaction Integrity	Weeks 2-3	Idempotency, @transaction.atomic, confirmation tracking, reconciliation
P2	Security Hardening	Weeks 3-4	Admin RBAC, KMS vault keys, HMAC signing, data masking
P3	Observability	Weeks 4-5	Sentry, structured logs, Prometheus, PagerDuty, audit trail
P4	Testing	Weeks 5-7	80%+ coverage on money paths, e2e, load tests
P5	Scalability	Weeks 7-9	PgBouncer, read replicas, Celery 3-worker, RPC failover
P6	Feature Completion	Weeks 9-11	Paystack/Flutterwave, notifications, referrals, NDPR export

Total: 14 weeks (3.5 months) with 1-2 backend engineers.

Phase 0 is non-negotiable before any real money flows. Phases 1-3 must complete before public launch. Phases 4-6 can run in parallel with a controlled beta.

12. Deployment & CI/CD

Service	Platform	Region	Scaling
clusteer-unified	Vercel	Global edge	Auto (serverless)
Blockchain Engine	Railway	US East	Vertical (Docker)
Spring Boot API	Railway	US East	Vertical (Docker)
PostgreSQL	Railway	US East	Managed
Redis	Railway	US East	Managed
Firebase	Google Cloud	us-central1	Auto (managed)

CI/CD Pipeline: GitHub Actions → Lint + Test + Security Scan → Vercel Preview (PR) / Vercel Production (main) + Railway auto-deploy (Docker).

13. Environment Variable Registry (Key Variables)

Variable	Service	Purpose
BLOCKCHAIN_ENGINE_URL	Next.js	Django backend URL
BLOCKCHAIN_ENGINE_API_KEY	Next.js	API key for Django
JWT_SECRET	Next.js	2FA token signing (32+ chars)
FIREBASE_PRIVATE_KEY	Next.js	Firebase Admin SDK
CUSTODIAL_WALLET_ENCRYPTION_KEY	Django	Encrypt user wallet keys
VAULT_WALLET_ENCRYPTION_KEY	Django	Encrypt vault keys
TATUM_API_KEY	Django	Deposit monitoring
HMAC_SECRET	Django	Webhook verification
BSC_RPC_URL / SOLANA_ENDPOINT / TRONS_PROVIDER	Blockchain	Chain RPC endpoints
PAYSTACK_SECRET_KEY	Next.js	Payment gateway
SENDGRID_API_KEY	Next.js	Email delivery

Repository: github.com/OutbuildHQ/clusteer_new

Branch: design-system-revamp (active) | Main: dev3